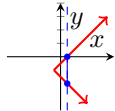


You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

Determine whether the following relation defines y as a function of x .

1. $x = |y + 3| - 1$ **No.** By the vertical line test.



Find the domain and range of the relation. Determine if the relation is a function.

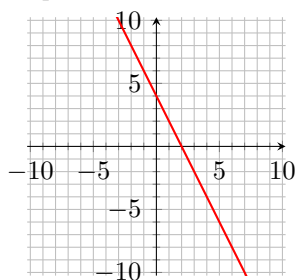
2. $g = \{(2, 7), (30, -4), (45, 7), (2, 5)\}$

- a. Domain: $\{2, 30, 45\}$
- b. Range: $\{7, -4, 5\}$
- c. Is g a function? **No.** $(2, 7)$ and $(2, 5)$ mean that g pairs 2 with 7 and 5.

Sketch the graph of the function, then give its domain and range.

3. $y = -2x + 4$

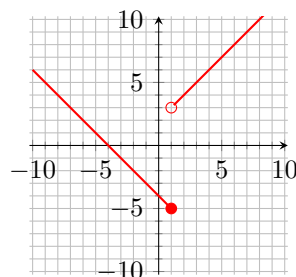
- a. Graph:



- b. Domain: $(-\infty, \infty)$, i.e., all real numbers
- c. Range: $(-\infty, \infty)$

4. $f(x) = \begin{cases} x + 2 & \text{for } x > 1 \\ -x - 4 & \text{for } x \leq 1 \end{cases}$

- a. Graph:

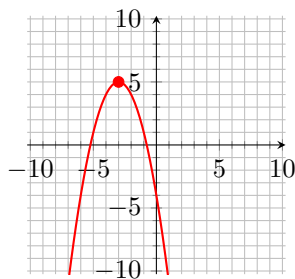


- b. Domain: $(-\infty, \infty)$
- c. Range: $[-5, \infty)$

Use transformation to graph the function. State the domain and range using interval notation.

5. $f(x) = -(x + 3)^2 + 5$

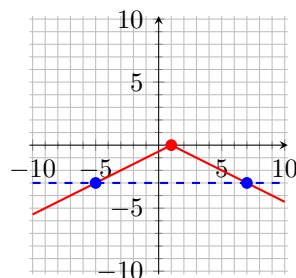
- a. Graph:



- b. Domain: $(-\infty, \infty)$
- c. Range: $(-\infty, 5]$

6. $g(x) = -\frac{1}{2}|x - 1|$

- a. Graph:

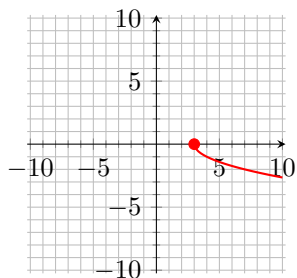


- b. Is $g(x)$ one-to-one? i.e., does it have an inverse? **No.** By the horizontal line test.

Sketch the graph and state the domain and range. Identify any intervals on which $f(x)$ is increasing or decreasing.

7. $f(x) = -\sqrt{x-3}$

a. Graph:



b. Increasing: $\emptyset$. It does not increase

c. Decreasing: $(3, \infty)$

Use the function $h(x) = 4x + 5$ to complete the following problems.

8. Find $h(4)$.

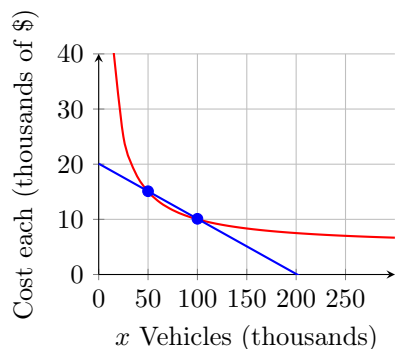
$$h(4) = 4(4) + 5 = 21$$

9. $h(0) = 5$. Find $h^{-1}(5)$

$$h^{-1}(5) = 0 \text{ since } h(0) = 5.$$

Use the provide graph to answer the question.

10. The manufacturing cost per vehicle for production of x vehicles is shown in the graph. Use the graph to find the average rate of change of the cost per vehicle from 50 to 100 thousand vehicles.



Using the points $(50, 15)$ and $(100, 10)$,

$$\begin{aligned} \text{Average rate of change} \\ \approx \frac{15 - 10}{50 - 100} = -\frac{1}{10} \end{aligned}$$

Use the difference quotient, $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$, to complete and simplify the following problem.

11. $f(x) = 7x + 3$.

$$\frac{[7(x+h) + 3] - [7x + 3]}{h} = \frac{7x + 7h - 7x - 3 + 3}{h} = \frac{7h}{h} = 7$$

Find the inverse of the following function.

12. $f(x) = -\frac{1}{3}x + 1$

$$\begin{aligned}y &= -\frac{1}{3}x + 1 \\y - 1 &= -\frac{1}{3}x \\-3(y - 1) &= x \\f^{-1}(x) &= -3(x - 1)\end{aligned}$$

Let $f(x) = x + 4$ and $g(x) = x^2 - x$. Find and simplify the expression.

13. $(f + g)(-6)$

$$\begin{aligned}g(-6) &= (-6)^2 - (-6) = 42 \text{ and } f(-6) = (-6) + 4 = -2. \\ \text{So, } (f + g)(-6) &= -2 + 42 = 40\end{aligned}$$

14. $(g \circ f)(2)$

$$\begin{aligned}f(2) &= 2 + 4 = 6 \text{ and } g(6) = 36 - 6 = 30. \\ \text{So, } g(f(2)) &= g(6) = 30\end{aligned}$$

For the given functions, find $(f \circ g)(x)$ and $(g \circ f)(x)$, Determine if g is the inverse of f .

15. $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$.

a. $(f \circ g)(x)$

$$f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 + 1 = x + 1$$

b. $(g \circ f)(x)$

$$g(f(x)) = g(x^2 + 1) = \sqrt{x^2 + 1}$$

c. Is g the inverse of f ? **No.** Either note that $f(g(x)) \neq x$ or $g(f(x)) \neq x$.